Door Status Indicator Module

AssignmentreportwithKiCaddesignevidenceandbillof materials

STUDENT
Shivanshi Pandey

PROJECT SCOPE
Schematic, PCB Layout, BOM

SUBMISSION FORMAT
KiCad Evidence & BOM Technical Report

01 Product Definition

Door Status Indicator Module: A 5 V PCB module designed for visual and audible door-status indication. The primary design objective is to create a robust 5 V PCB module that accurately indicates door status using a reed switch sensor, red and green indicator LEDs, an active buzzer, current-limiting resistors, and a dedicated power connector.

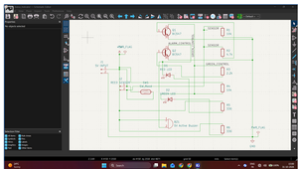
01
Door Open
RedLED indication paired with audible alert control.

02
Door Closed
Green LEDindication signaling secure state.

03
Design Tool
KiCadEDAsuite (Schematic & PCB Editor).

02 Schematic Design

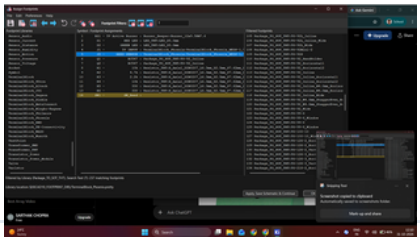
Design evidence showcasing the complete KiCad schematic structure including power input, reed sensor interface, transistor switching logic, dual LED indicators, active buzzer drive, power flag, and ground connections.



Circuit logic overview: J1 acts as the primary 5 V DC power input; connector J2 and SW1 form the magnetic reed sensor circuit. D1 serves as the red LED status indicator and D2 as the green LED. NPN Transistors Q1 and Q2 (BC547) drive the active buzzer alarm logic and green control switching path respectively.

03 Footprint Assignment

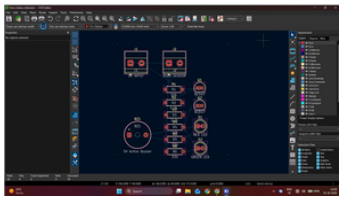
Footprint mapping and verification for the complete component library set prior to PCB component placement and trace routing.



Selected through-hole component footprints include: 2-pin screw terminal blocks for J1–J2, inline reed switch package SW1, 5 mm standard through-hole LEDs for D1–D2, 12 mm active buzzer BZ1, TO-92 package NPN transistors Q1–Q2, and axial DIN0207 resistors R1–R6.

04 PCB Placement and Routing

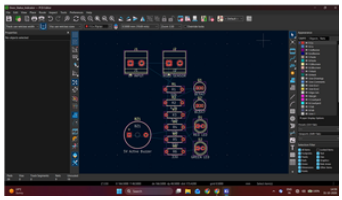
Compact through-hole layout strategy: primary power connector and reed sensor interface positioned at top edge, visual indicators grouped on the right side, and active buzzer located at the lower left quadrant.



Routing verification checklist: Electrical clearance checks, power return paths, physical connector access validation, and explicit component polarity orientation marks verified prior to final manufacturing release.

05 2D PCB View After Routing

Fully routed 2D printed circuit board top-layer view depicting component footprints, silk-screen layers, and track routing structure.



06 3D PCB Visual Rendering

Three-dimensional mechanical visualization of the populated PCB assembly.



The 3D model confirms component clearance and physical layout for two terminal blocks, two LEDs, two TO-92 transistors, six axial resistors, and one active buzzer. *Note: Verify Edge.Cuts board outline definition prior to Gerber generation.*

07 Bill of Materials (BOM)

Complete component breakdown transcribed from schematic and footprint evidence.

Ref	Qty	Value / Description	Footprint / Package
J1	1	5 V Input Terminal	2-pin Terminal Block
J2	1	Reed Sensor Connector	2-pin Terminal Block
SW1	1	SW_Reed	Reed Switch
D1	1	RED LED	5 mm Through-Hole LED
D2	1	GREEN LED	5 mm Through-Hole LED
BZ1	1	5 V Active Buzzer	Buzzer_Beeper: Buzzer_12.5x7.6_P7.6mm
Q1–Q2	2	BC547 NPN Transistor	TO-92 Inline
R1	1	10 kΩ Resistor	Axial DIN0207 L6.3mm D2.5mm P7.62mm
R2	1	4.7 kΩ Resistor	Axial DIN0207 L6.3mm D2.5mm P7.62mm
R3	1	2.2 kΩ Resistor	Axial DIN0207 L6.3mm D2.5mm P7.62mm
R4	1	10 kΩ Resistor	Axial DIN0207 L6.3mm D2.5mm P7.62mm
R5–R6	2	330 Ω Resistor	Axial DIN0207 L6.3mm D2.5mm P7.62mm

08 Submission Checklist & Repository Link

- [ ] **Student Name:** Shivanshi Pandey
- [ ] **Project Title:** Door Status Indicator Module
- [ ] Schematic screenshot included
- [ ] Footprint assignment and routed PCB evidence included
- [ ] 2D PCB view included
- [ ] 3D PCB view included
- [ ] Bill of Materials (BOM) included
- [ ] **GitHub Repository:** https://github.com/Shivanshi2810/Shivanshi

Note: Upload the PDF report and Gerber files to the GitHub repository above. Paste the same GitHub link into a Word document with the student name, convert to PDF, and submit on the LMS portal.